RESP-100 · BASIC EKG INTERPRETATION · WEEK 10

Week 10 Study Guide: Ventricular Rhythms

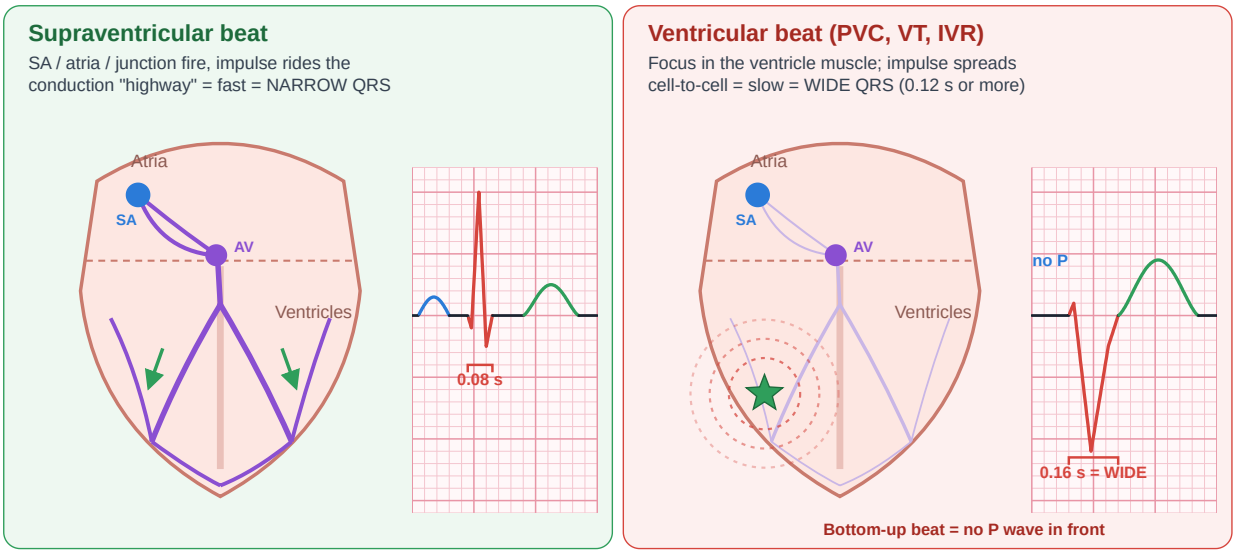
Walraven, *Basic Arrhythmias* 8th ed. · Chapter 8 · Lecture by Michael V Hyland (1 h 11 min)

In one line: when the ventricles make the beat, the QRS is **wide (0.12 s or more)**, there is **no P wave**, and R and T point in **opposite directions**. Learn the PVC patterns, then the four life-threatening rhythms.

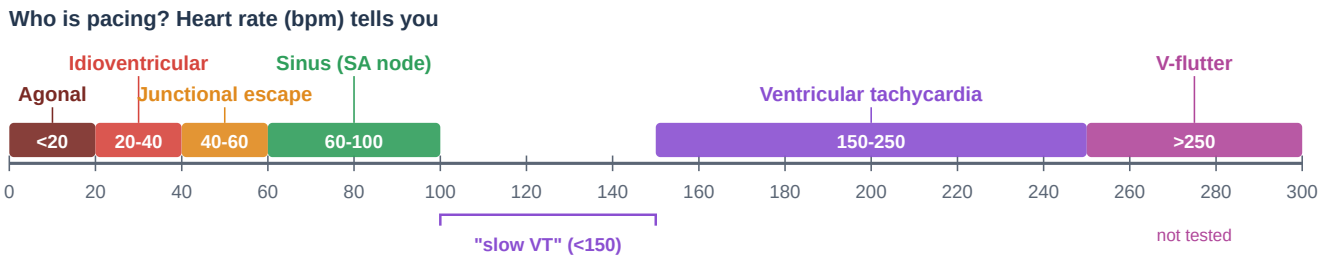
1 · Why ventricular beats look wide | 2 · Anatomy of a PVC | 3 · Measuring a wide QRS | 4 · PVC patterns & names | 5 · Worked example | 6 · Ventricular tachycardia | 7 · Ventricular fibrillation | 8 · Idioventricular & agonal | 9 · Asystole | 10 · What to do | Cheat sheet | Quiz

1 Why ventricular beats look wide

Big idea: a beat from above the ventricles rides the conduction "highway" and is fast, so the QRS is narrow. A beat that starts in ventricular muscle spreads **cell to cell**, slowly, so the QRS is drawn **wide and bizarre**, and since it fires from the bottom up there is **no P wave** before it.



Left: normal (supraventricular) conduction, narrow QRS. Right: ventricular focus (green star) spreading slowly cell-to-cell, wide QRS, no P.



Escape (backup) pacemakers fire at their inherent rate; irritable foci fire FAST and take over ("fastest rate wins").

The rates to memorise. The ventricles' own (escape) rate is 20-40. An *irritable* ventricular focus fires fast (VT 150-250).

Two ways the ventricles take over

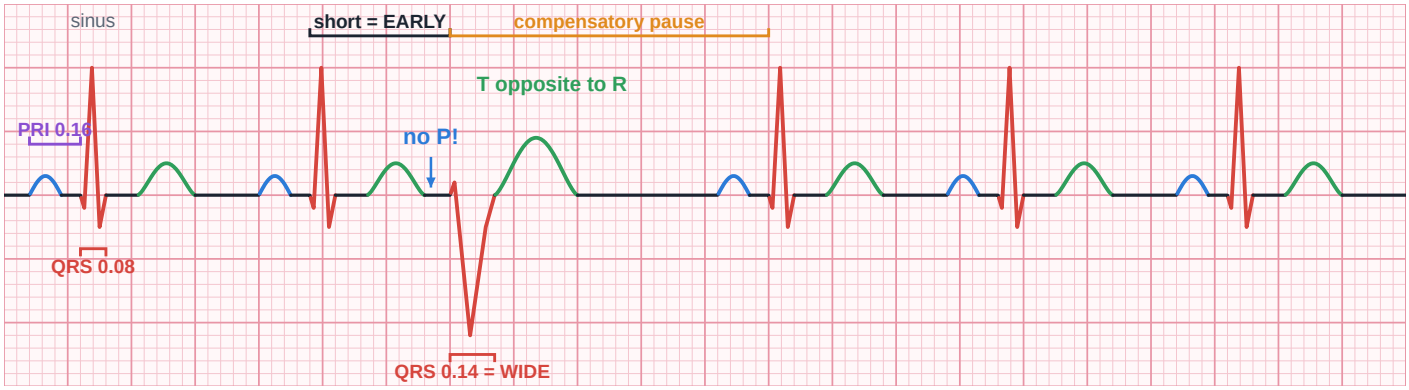
- **Irritability** (early, fast): PVCs, runs, **VT**, **VF**. Fastest rate wins.
- **Escape** (backup): SA node *and* AV node fail, so the ventricles pace at 20-40: **idioventricular rhythm** (ventricular escape rhythm, VER). The *last* backup pacemaker.

WHAT THE INSTRUCTOR STRESSED (0:00, 0:42)

Ventricular rhythms and heart blocks are "the two most challenging topics" of the semester. "None of the rhythms I've covered so far are life-threatening, but the ones I'm going to get into next are … this chapter is super important."

2 Anatomy of a PVC (premature ventricular complex)

Big idea: a PVC is a **single early beat** from one irritable spot in the ventricles. It is *not* a rhythm, so you must still name the **underlying rhythm** (just like PACs and PJCs).

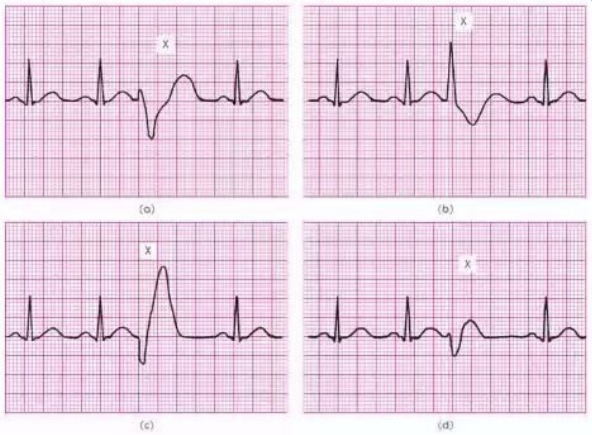


Normal sinus rhythm (rate 83) with one PVC. Colours: P QRS T PRI.

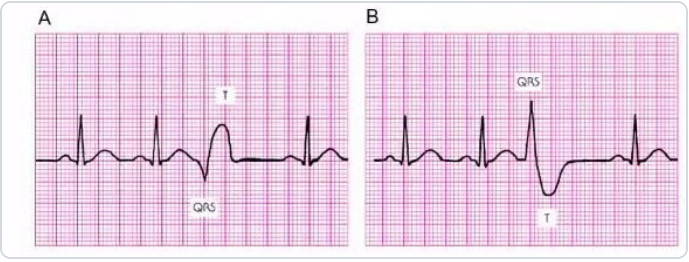
The 4 tell-tale signs of a PVC

1. **Early**: it comes too close to the beat before and breaks the regular spacing.
2. **No P wave** in front (the beat starts at the bottom), so **no PRI**.
3. **Wide & bizarre QRS**: 0.12 s or more (3+ small boxes).
4. **R and T go opposite ways** ("98-99% of the time").

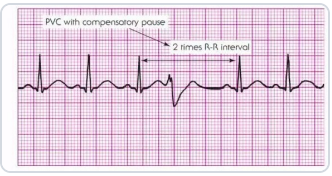
Usually followed by a **compensatory pause**.



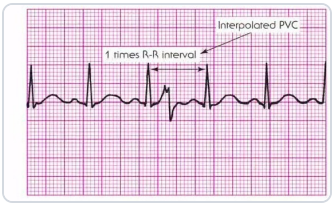
PVCs can point up, down or both, depending on which electrode the impulse travels towards. Shape doesn't matter; the 4 signs do.



Slide: T-wave configuration. A: R down, T up. B: R up, T down.



Compensatory pause: about 2 × R-R.



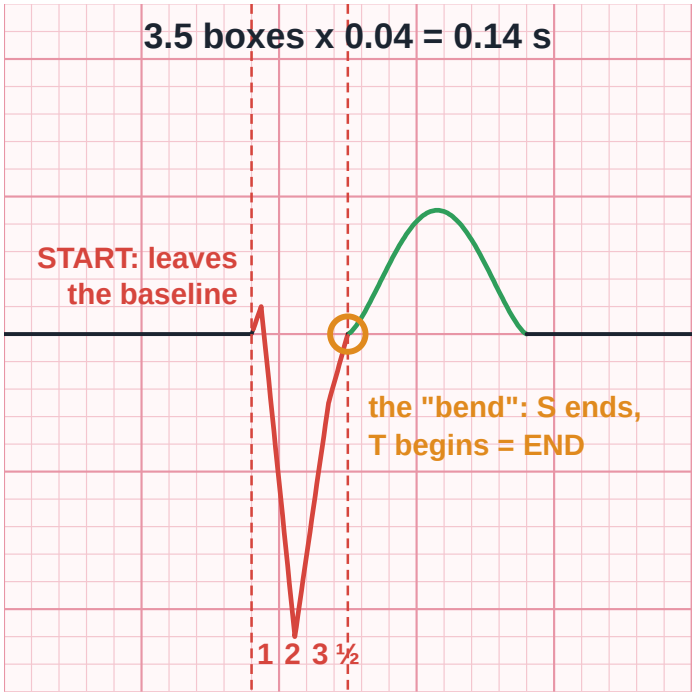
Interpolated PVC: no pause.

NOT TESTED (10:12)

The pause and the word "interpolated" won't be on the quiz. What matters: **recognise that the early beat is a PVC.**

3 Measuring a wide QRS: "find the bend"

Big idea: on a PVC it's hard to see where the S wave ends and the T begins. Look for the **bend** (little hitch or notch) where the downstroke turns into the T wave. Measure from where the complex first leaves the baseline **to the bend**. Don't include the T wave.



Zoomed PVC: count the small boxes from start to bend. 3.5 boxes = 0.14 s.

- He often **counts backwards**: find a clear end-point on a grid line, then count back to the start.
- No clear bend? Run straight across at the **isoelectric line** and measure the width there.
- Half boxes count: 3.5 boxes = 0.14 s; 7.5 boxes = 0.30 s.
- He gives a **1-box grace** on measurements.

GRADING RULE (WEEK 11 REVIEW, 0:19:59)

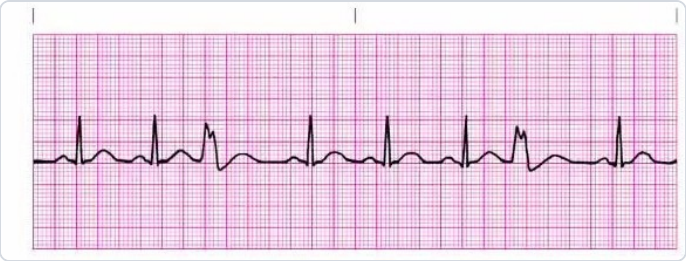
On the QRS line write **both** widths: **underlying / PVC**, e.g. **0.08 / 0.16**. If two multifocal PVCs have different widths, give a third number. Leaving one out costs a point.

4 PVC patterns: the names you must use

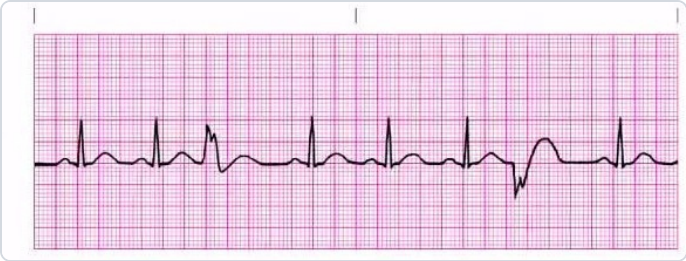
Big idea: after "NSR with …" you must name **how** the PVCs appear. "NSR with 2 PVCs" loses a point. The instructor wants "unifocal" or "multifocal", "couplet", "run of VT", or "bigeminy / trigeminy / quadrigeminy".

Unifocal PVCs		PVCs look the SAME (one focus)
Multifocal PVCs		PVCs look DIFFERENT (2+ foci)
Couplet		2 PVCs in a row ("holding hands")
Run / burst of VT		3 or more PVCs in a row
Bigeminy		PVC every 2nd beat
Trigeminy		PVC every 3rd beat
Quadrigeminy		PVC every 4th beat

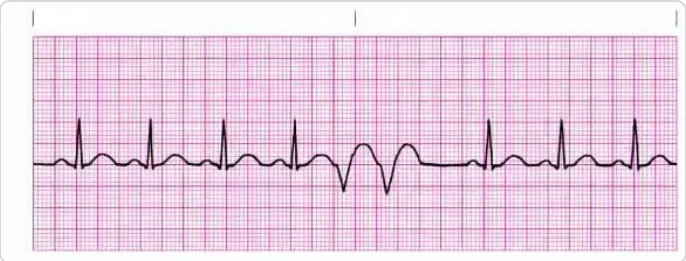
Pattern pictograms: black = normal (narrow) beat, red/purple = PVC.



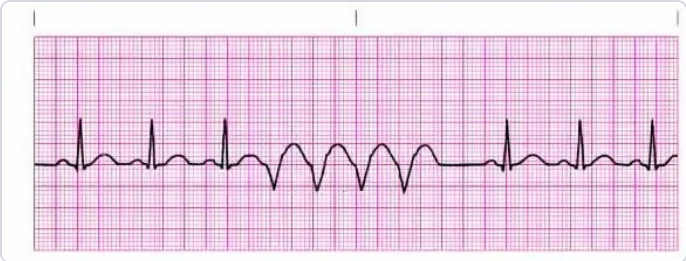
Unifocal: beats 3 and 7 look identical (same focus). NSR with 2 unifocal PVCs.



Multifocal: one PVC points up, one down (2 foci). NSR with 2 multifocal PVCs.

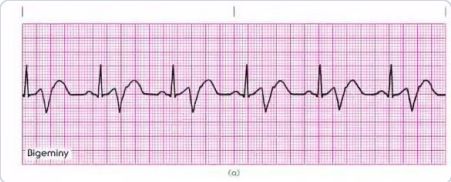


Couplet: 2 PVCs back to back. NSR with one couplet of PVCs.

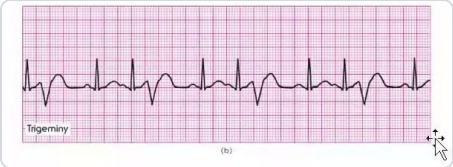


Run: 3 or more in a row = run / burst of VT. NSR with a 4-beat run of VT.

Grouped beating: bigeminy, trigeminy, quadrigeminy



Bigeminy (every 2nd beat). R-R 26 boxes, rate 58: **sinus bradycardia with bigeminy**.



Trigeminy (every 3rd). R-R 13 boxes, rate 115: **sinus tachycardia with trigeminy**.



Quadrigeminy (every 4th). Rate 115: **sinus tachycardia with quadrigeminy**.

MEMORY TRICK

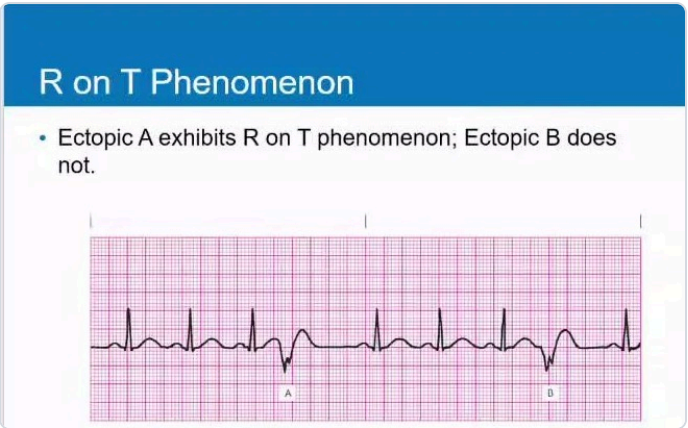
Bi = 2 (bicycle): PVC is every **2nd** beat. **Tri** = 3 (tricycle): every **3rd**. **Quad** = 4: every **4th**. Count the PVC itself as one of the beats: N-V, N-N-V, N-N-N-V.

RATE TRAP

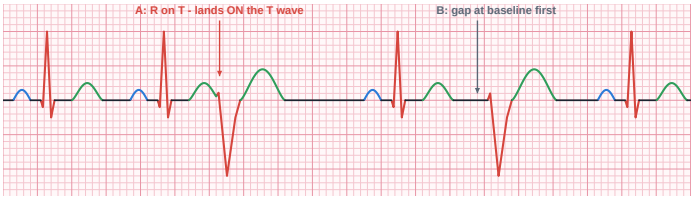
Never count the PVCs in the heart rate. They may not eject blood, so they can't be relied on for cardiac output. Measure R-R between the **underlying** beats (1500 ÷ boxes). If the underlying rhythm is irregular, count only the **narrow** complexes in 6 s × 10.

REGULARITY RULE OF THUMB

If you can find **two R-R intervals of the underlying rhythm that are within 1.5 boxes** of each other, the underlying rhythm is **regular**. Write: **"regular with ectopics"** (e.g. "regular with an ectopic every other beat").



R on T: ectopic A lands on the T wave; B doesn't.



A PVC landing on the T wave hits the ventricles while they are repolarising.

R ON T: WHY IT MATTERS (NOT TESTED)

Can "trick" the heart into **VT or VF**. If the underlying rate is adequate, PVCs may be suppressed with anti-arrhythmic drugs.

5 Worked example: the instructor's 5-step method

Big idea: gather the 5 pieces of information for the **underlying rhythm first**, then investigate the early beats, then put the name together.

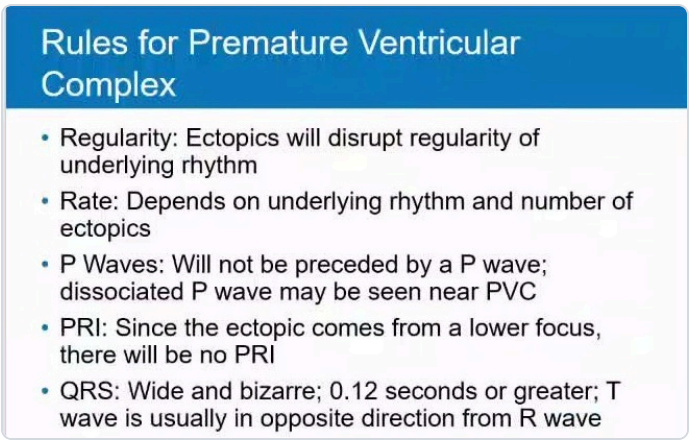


Unifocal PVC strip, worked through in lecture at 0:12:31.

Step	What he did	Result
1. Regularity	Counted R-R: 17.5, 18, 18 … early beats break it (11.5, then a long pause of 24.5).	Regular with 2 ectopics
2. Rate	Underlying spacing 18 boxes: 1500 ÷ 18	83 bpm
3. P waves	Underlying beats: upright and all alike. Early beats: none.	Upright & uniform
4. PRI	4 small boxes on every underlying beat	0.16 s
5. QRS	Underlying 2 boxes; early beats 4 boxes	0.08 / 0.16
Name it	Underlying is all normal = sinus. Early + no P + wide = PVC. Both identical = unifocal.	**NSR with 2 unifocal PVCs**

Rules for PVCs

Premature Ventricular Complex	
Regularity	Depends on the underlying rhythm; the ectopics interrupt it
Rate	Of the underlying rhythm (PVCs not counted)
P wave	None before the PVC (assess the underlying P waves)
PRI	None for the PVC (measure the underlying PRI)
QRS	Wide & bizarre, 0.12 s or more; T usually opposite R



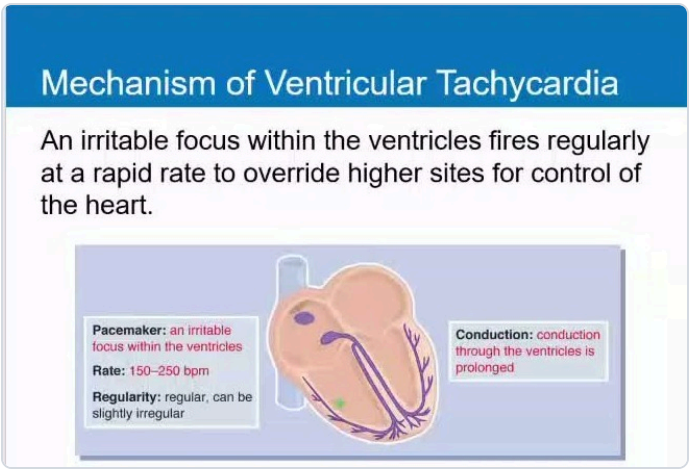
The textbook rules slide.

PVCS CAN SIT ON ANY RHYTHM

You could see a PVC in atrial fib or a junctional rhythm too. Always name the underlying rhythm first, then "with … PVCs".

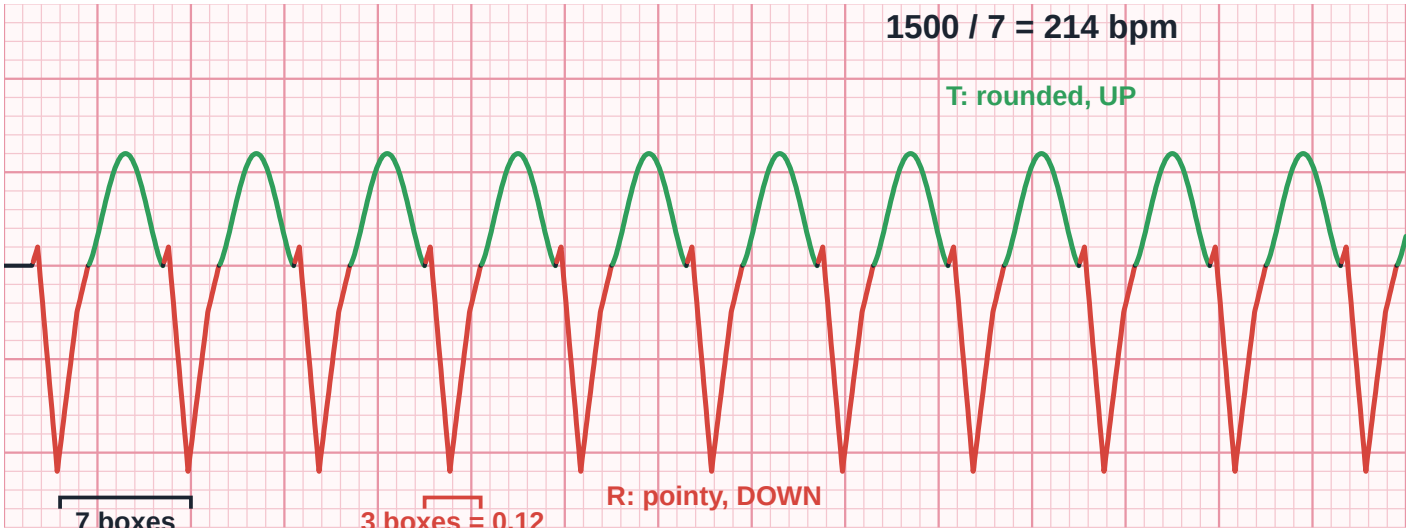
6 Ventricular tachycardia (VT, "V-tach")

Big idea: the same irritable ventricular focus now fires **repetitively and fast (150-250)**, faster than the SA node, so it takes over. VT is PVC after PVC after PVC. It's **immediately life-threatening**.

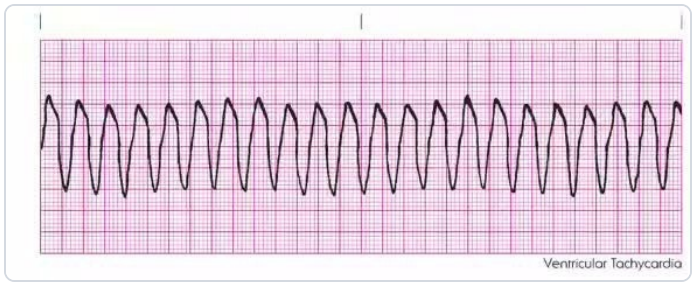


Mechanism: one irritable focus firing rapidly.

Ventricular Tachycardia	
Regularity	Usually regular (can be slightly irregular)
Rate	150-250. Below 150 = "slow VT". Above 250 = ventricular flutter (never tested)
P wave	None
PRI	None
QRS	Wide & bizarre, 0.12 s or more; T opposite R



Reading VT like the instructor: the **pointier** waves are the R waves (down), the **rounder** ones are T waves (up). R-R 7 boxes gives 1500 ÷ 7 = 214 bpm. QRS from the bend to the bend: 3 boxes = 0.12 s.



Lecture strip (0:53:54): regular, 214 bpm, no P, QRS 0.12: **ventricular tachycardia**.

WHY IT KILLS

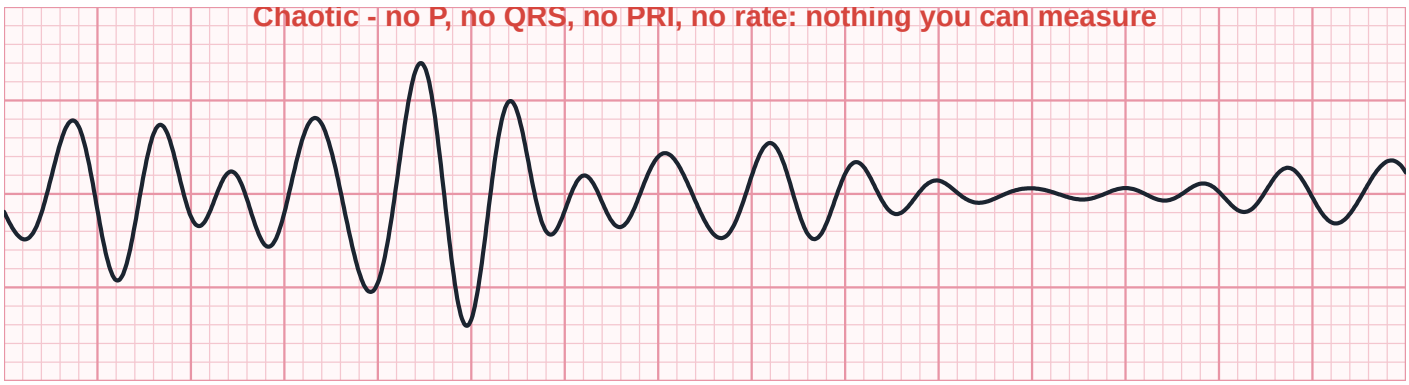
At 214 bpm that's about 3.5 contractions per second, so the ventricles have **no time to fill**. Usually **pulseless**. Even with a pulse, it's weak with a low BP. **Shockable** rhythm.

EXAM ALERT (0:57:56)

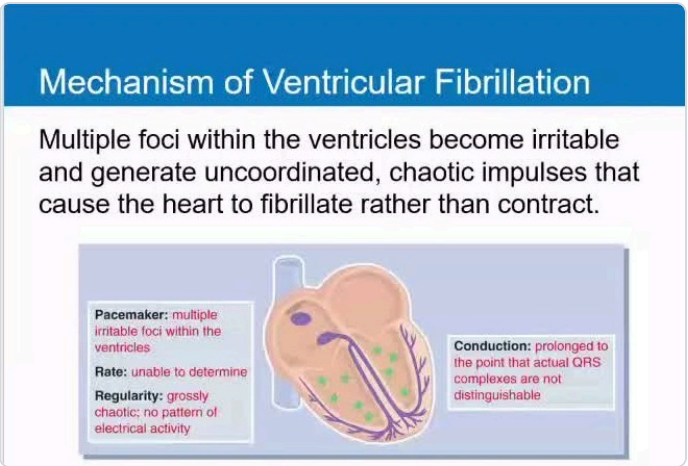
"You're going to see this on the quiz, … on the next hourly exam, … on the final exam."

7 Ventricular fibrillation (VF, "V-fib")

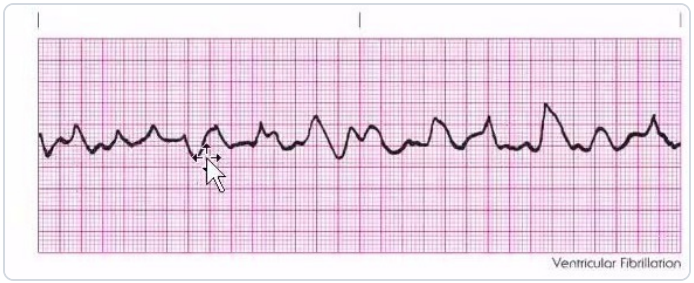
Big idea: **many** irritable foci fire at once. The ventricles can't respond to them all, so they **quiver instead of contracting**: no cardiac output, no pulse, no BP. It's like A-fib, but in the ventricles.



Your words for VF: Regularity **chaotic**; Rate **none**; P **none**; PRI **none**; QRS **none**.



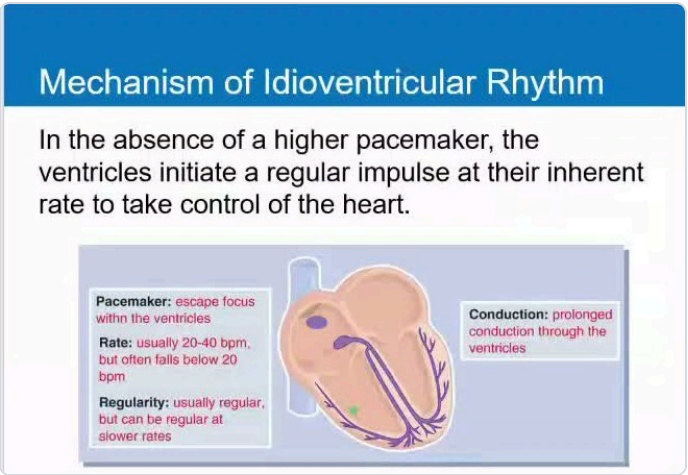
Mechanism: multiple irritable foci (green stars).



Lecture strip: no recognisable waveforms, so it's ventricular fibrillation.

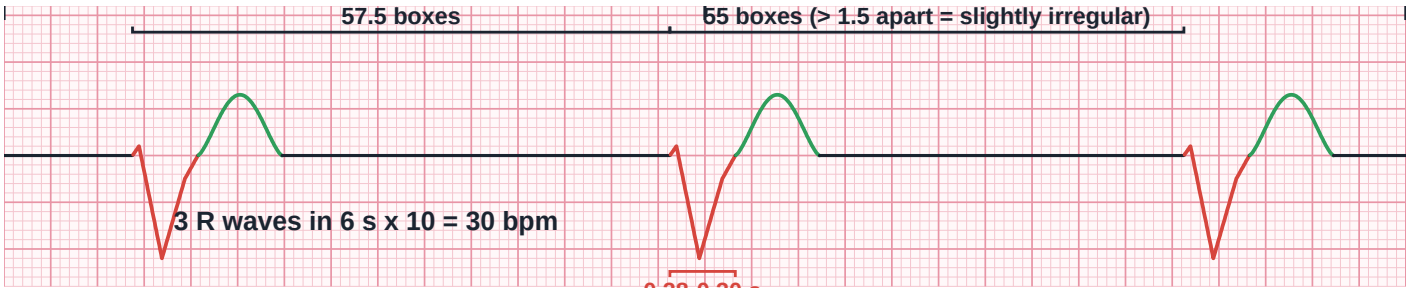
8 Idioventricular rhythm (ventricular escape) & agonal rhythm

Big idea: not irritability but **escape**. The SA node fails, then the AV node fails, so a spot in the ventricles becomes the **last backup pacemaker** at its inherent rate of **20-40**. That's too slow to keep up a blood pressure, so it's **life-threatening**.

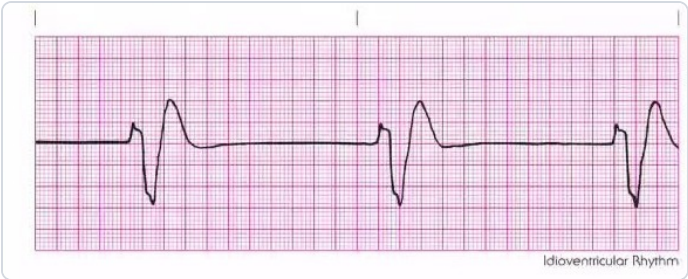


Mechanism: escape focus in the ventricles.

Idioventricular Rhythm (VER)	
Regularity	Usually regular
Rate	20-40 (below 20 = **agonal**)
P wave	None
PRI	None
QRS	Wide & bizarre, 0.12 s or more; T opposite R



Lecture example (1:03:42): spacing 57.5 and 55 boxes (more than 1.5 apart, so slightly irregular). Count R waves × 10 = 30 bpm. QRS 7.5 boxes = 0.30 s. Interpretation: **idioventricular rhythm** (or ventricular escape rhythm).



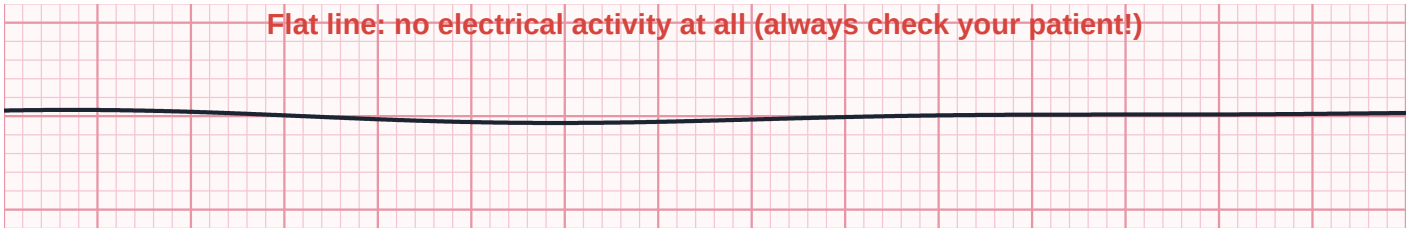
Lecture strip: the first bump is part of the QRS, *not* a P wave.

AGONAL RHYTHM = "DYING HEART"

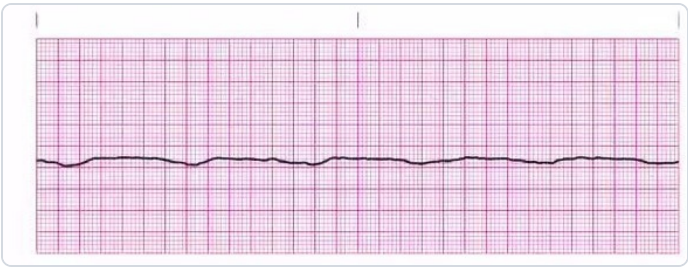
An idioventricular rhythm that slows to **under 20**. It's a terminal, lethal rhythm: the heart has stopped beating in a reliable pattern. (Agonal breathing = gasping.) He won't give you a VER under 20 to interpret.

9 Asystole ("flat line")

Big idea: the prefix **a-** means "without", so asystole is **without systole**: no electrical activity, no contraction, no pulse. No rate, no P, no PRI, no QRS.



Asystole: essentially a flat line. Not shockable: call 911 and start CPR (the AED will say "no shock advised").

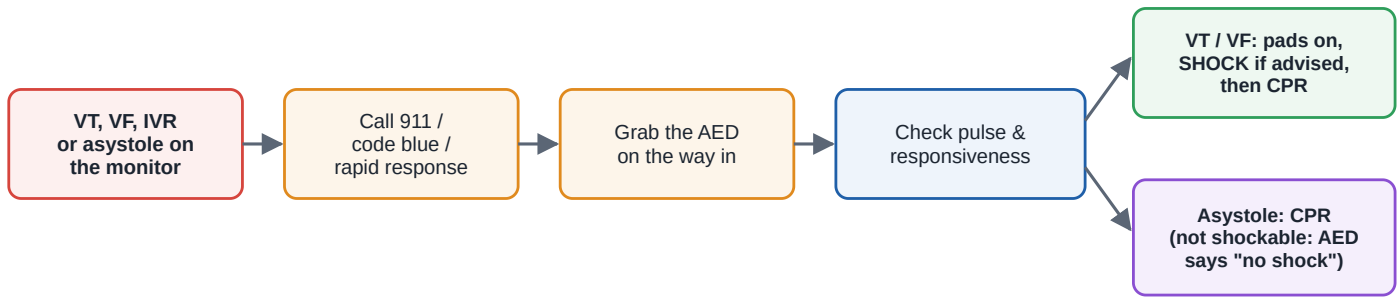


Lecture strip: asystole.

BONUS: PEA (WEEK 11 REVIEW)

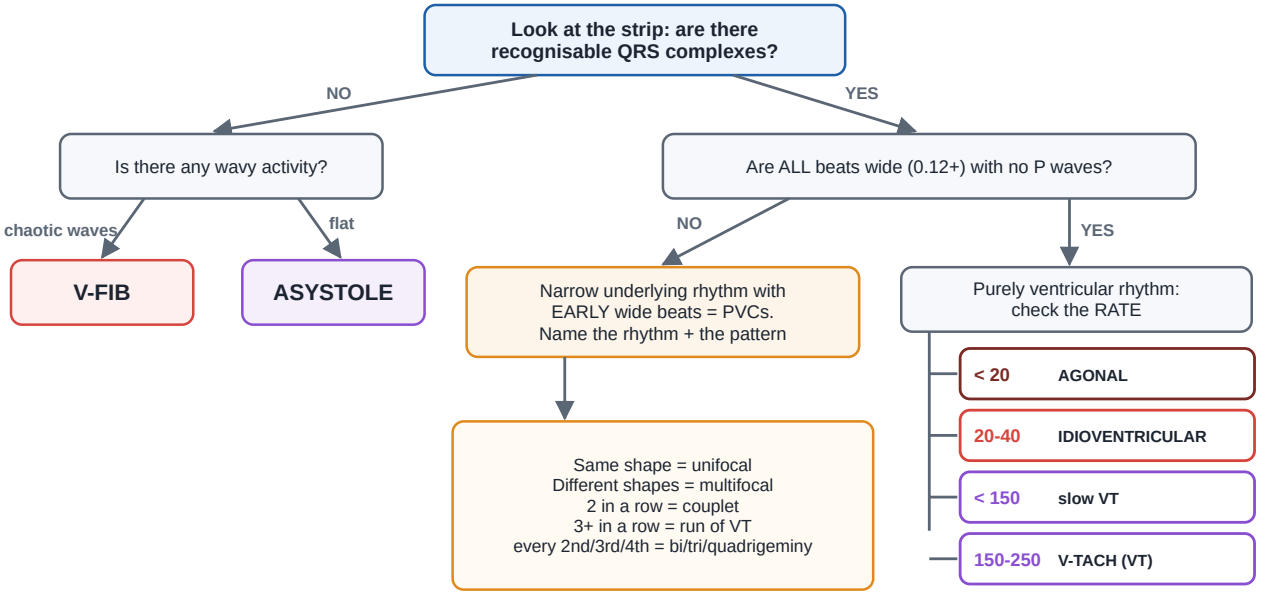
You can have **electrical activity without contraction**: a normal-looking rhythm on the monitor but a blue, pulseless patient. That's **pulseless electrical activity (PEA)**, a code situation. (You can't have contraction without electrical activity.) Not on homework or tests.

10 What to do: the life-threatening four



The instructor's sleep-lab scenario: pick up the phone, grab the AED, go in. You'll learn the full steps in BLS.

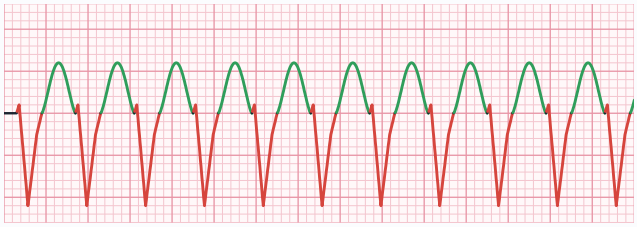
Putting it together: which ventricular rhythm is it?



Decision tree for any strip with wide or missing QRS complexes.

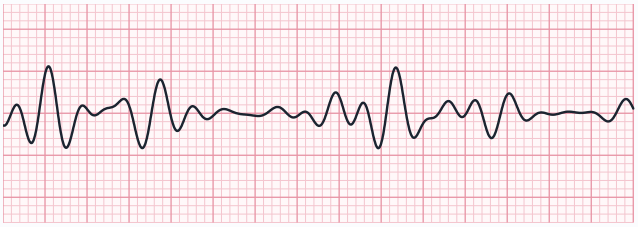
The life-threatening four at a glance

Ventricular tachycardia



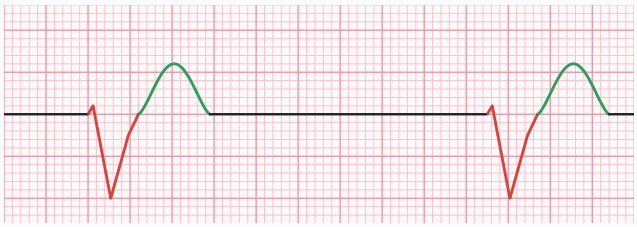
Fast (150-250), regular, wide, no P. Looks like a row of PVCs. **Shock.**

Ventricular fibrillation



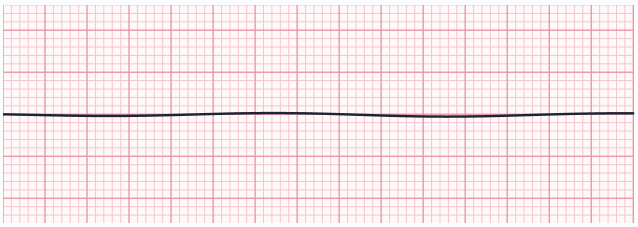
Chaotic squiggles, nothing to measure. **Shock.**

Idioventricular (VER)



Slow (20-40), wide, no P, long flat gaps. Below 20 = agonal.

Asystole



Flat line. **CPR**, not shockable.

★ Cheat sheet: Chapter 8 on one page

Rhythm	Regularity	Rate	P wave	PRI	QRS	Key clue
PVC (single beat)	Of underlying; ectopic interrupts	Of underlying (don't count PVCs)	None on PVC	None on PVC	≥0.12, bizarre, T opposite R	Early + no P + wide
Ventricular tachycardia	Usually regular	150-250	None	None	≥0.12, T opposite R	PVC after PVC; shockable
Ventricular fibrillation	**Chaotic**	None	None	None	None	No recognisable waves; shockable
Idioventricular (VER)	Usually regular	20-40	None	None	≥0.12, T opposite R	Slow, wide, last backup
Agonal	Unreliable	<20	None	None	Wide	Dying heart
Asystole	None	None	None	None	None	Flat line; not shockable

PVC term	Means	Example answer
Unifocal	PVCs look the same	NSR with 2 unifocal PVCs
Multifocal	PVCs look different	NSR with 2 multifocal PVCs
Couplet	2 in a row	NSR with one couplet of PVCs
Run / burst of VT	3+ in a row	NSR with a 4-beat run of VT
Bigeminy	Every 2nd beat	Sinus brady with bigeminy
Trigeminy	Every 3rd beat	Sinus tach with trigeminy
Quadrigeminy	Every 4th beat	Sinus tach with quadrigeminy

HOW TO WRITE YOUR ANSWER

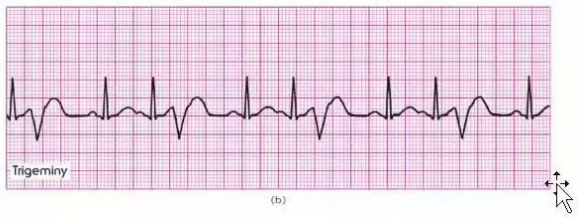
Regularity: "regular with 2 ectopics" / "… with an ectopic every other beat"
Rate: underlying only
P: underlying "upright & uniform"
PRI: underlying, e.g. 0.16
QRS: underlying / PVC, e.g. 0.08 / 0.14
Interpretation: underlying rhythm + PVC term

NUMBERS TO OWN

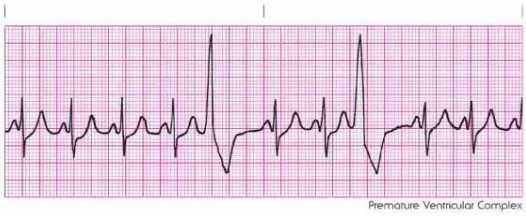
QRS wide = **≥0.12 s (3 boxes)** · ventricles' own rate **20-40** · junction 40-60 · SA 60-100 · VT **150-250** · agonal **<20** · run of VT = **3+** PVCs

? Self-quiz

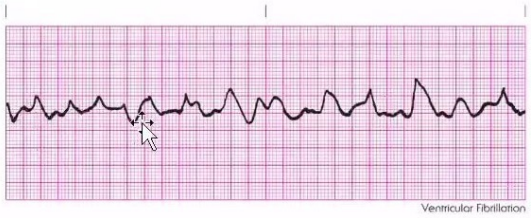
1. Give the 4 features that tell you an early beat is a PVC.
2. Why is the QRS of a ventricular beat wide? Why is there no P wave?
3. A strip has PVCs that look different from each other. What word must you add to the interpretation?
4. What is the name for 2 PVCs in a row? For 3 or more in a row?
5. Why don't we count PVCs when calculating the heart rate?
6. Measure: a PVC starts on a grid line and its bend is 3.5 small boxes later. What do you write on the QRS line if the underlying QRS is 2 boxes?
7. Identify this rhythm. The underlying R-R is 13 boxes.



8. Identify this rhythm (R-R of underlying beats is 14 boxes; PRI 3 boxes; QRS 2 boxes).



9. In VT, how do you tell the R waves from the T waves? A VT strip has R-R = 7 boxes: what is the rate?
10. Identify this rhythm and state what you would do first.



11. SA and AV node both fail. Which rhythm appears, at what rate, and why is it life-threatening?
12. What is an agonal rhythm?
13. What does R on T mean and why is it dangerous?
14. Which two of the life-threatening rhythms are shockable, and which one is not?

Answers

1. Early (breaks the spacing); no P wave (so no PRI); wide & bizarre QRS ≥ 0.12 s; R and T in opposite directions.
2. The impulse travels cell to cell through muscle instead of along the fast conduction pathway, so depolarisation is slower and the QRS is drawn wider. The beat starts in the ventricles (bottom up), so there is no atrial P wave in front.
3. **Multifocal** (more than one irritable focus). Same-looking = unifocal. Leaving it out costs a point.
4. A **couplet**; 3+ = a **run (burst) of VT**.
5. We don't know if a PVC actually ejects blood (perfuses), so it can't be counted on for cardiac output.
6. 3.5 × 0.04 = 0.14, so write **0.08 / 0.14**.
7. 1500 ÷ 13 = 115, PVC every 3rd beat: **sinus tachycardia with trigeminy**.
8. 1500 ÷ 14 = 107; PRI 0.12; QRS 0.08 / 0.14; two identical PVCs: **sinus tachycardia with 2 unifocal PVCs**.
9. R waves are pointier (here negative), T waves rounder (here positive). 1500 ÷ 7 = **214 bpm**.
10. **Ventricular fibrillation** (chaotic, no rate/P/PRI/QRS). Call 911 / code blue, get the AED, check pulse, pads on and shock, CPR.
11. **Idioventricular (ventricular escape) rhythm**, 20-40 bpm: too slow to maintain cardiac output / blood pressure.
12. A ventricular escape rhythm slowed to **<20**: a terminal "dying heart" rhythm.
13. The PVC's R wave lands on the previous T wave, during ventricular repolarisation, which can trigger VT or VF.
14. VT and VF are shockable; asystole is not (CPR).